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Review Article

Digital image watermarking using deep learning: A survey

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ABSTRACT

Lately, a lot of attention has been paid to securing the ownership rights of digital images. The expanding usage of the Internet causes several problems, including data piracy and data tampering. Image watermarking is a typical method of protecting an image's copyright. Robust watermarking for digital images is a process of embedding watermarks on the cover image and extracting them correctly under different attacks. The embedded watermark might be either visible or invisible. Deep learning extracts image features using neural networks, which are highly effective in feature extraction. Watermarking techniques that utilize deep learning have gained a lot of interest due to their remarkable ability to extract features. This article offers an overview of digital image watermarking and deep learning. This article will discuss several research articles on digital image watermarking in deep-learning environments.

1. Introduction

Digital image watermarking has received more attention lately because of the rise in copyright disputes [1]. The process of hiding a watermark in images to preserve its copyright is known as digital image watermarking [2]. There are two watermarking processes: spatial domain and frequency domain [3]. Spatial domain techniques include patchwork [4], correlation, spread spectrum [5], and least significant bit (LSB) [6]. The cover image's pixel values are directly altered during the watermarking embedding process in the spatial domain. In contrast, image pixels are converted to frequency coefficients in the frequency domain and then altered [7]. Techniques in the transform domain include singular value decomposition (SVD) [8], discrete cosine transform (DCT) [9], discrete wavelet transform (DWT) [10], and others. Despite being computationally easier to implement, spatial domain watermarking provides less protection against geometric assaults. However, watermarking in the transform domain became more robust to attacks [11].

Watermarks can be categorized as visible or invisible [12]. Visible watermarks are clear and embossed on the cover media. However, their limited imperceptibility allows for visual identification [13]. Invisible watermarking embeds hidden data into cover media that authorized users can only identify through specific logical processes. However, it may cause an overhead in cover media [14]. The invisible watermarking methods are categorized into robust, fragile, and semi-fragile [15]. Robust watermarking protects copyright and identifies ownership since it is designed to withstand attempts to erase or destroy the mark [16]. Small modifications can easily affect the fragile watermark [17]. Semi-fragile approaches protect concealed data from purposeful attacks while making it vulnerable to malicious attacks [18].

Watermarking techniques can be blind, non-blind, or semi-blind based on the embedding and extraction process [19]. Blind watermarking extracts a watermark without knowing the original information [20]. Semi-blind watermarking does not need the original data to extract the watermark; it only needs the secret key and the watermark [21]. In non-blind watermarking, the watermark must be extracted with knowledge of the cover image and the secret key [22]. Fig. 1 summarizes the image watermarking categorization based on human perception, transform domain, and procedure type. Traditional watermarking approaches protect picture copyright by embedding an invisible but detectable signal into the host image. Therefore, watermarked pictures generated using these methods are always distorted [23]. Zero watermarking adds no data to the cover picture itself. Instead, it generates an ownership authentication key using a watermark signal and the cover picture's selected features [24].

The fundamental steps in watermarking are illustrated in Fig. 2. Using a secret key, the secret data is embedded into the cover image during the embedding procedure. It is accomplished by applying spatial or transform domain approaches [25]. Upon receiving the watermarked image, the receiver proceeds to the extraction step, which is the last phase of the process. The extracted watermark is recovered by following the extraction procedure [26].

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Important characteristics of the watermarking technique are measured using a variety of performance metrics, such as bit error rate (BER), peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM) [27]. The watermarking approach has several noteworthy uses, as seen in Fig. 3. These applications include military, e-voting, copyright protection, tamper detection, etc. The primary objectives of the watermarking technique are robustness, security, and imperceptibility.

The robustness of traditional watermarking algorithms is limited, and it is frequently challenging to identify the best embedding areas, as these algorithms adopt the manual method of embedding the watermark [28]. Neural networks, which are excellent at extracting features, have been included in digital watermarking to enhance their robustness compared to most conventional techniques [29].

Watermarking with deep learning has recently gained significant interest due to its numerous widely used applications and significant impact on the security of image content [30]. Deep learning has several benefits for watermarking images since the best embedding positions in the cover image may be found, and the ideal embedding strength can be determined while maintaining the robustness and imperceptibility of the image. Deep learning can decrease noise in the final watermark and efficiently simulate attacks to extract watermarks [31].

Deep learning is a branch of artificial intelligence that uses neural networks to simulate how the human brain processes information and makes decisions [32]. Deep learning extracts image features using neural networks and adaptively embeds the watermarking to provide superior outcomes [33]. Deep learning-based watermarking technology research is still in its early phases. Even if there aren't many studies, they appear to have become more numerous recently [34]. Numerous deep learning models exist, such as generative adversarial networks (GAN), deep neural networks (DNN), and convolutional neural networks (CNN) [35]. Based on the utilized network architecture, deep learning can consist of convolutions, fully connected layers, pooling layers, gates, etc. [36]. With the help of deep learning platforms, users can easily construct, train, and implement strong neural networks for various uses without starting from scratch, for instance, Keras, Torch, and Tensor Flow [37].

This paper provides a broad review of deep learning-based digital image watermarking and outlines some recent research publications. The remaining sections of this work are arranged as follows: An overview of the most popular deep learning models and their recent research publications is provided in Section 2. A discussion is provided in Section 3. Future directions in deep learning-based image watermarking techniques are provided in Section 4. Section 5 concludes with some findings.

2. Popular deep learning models based image watermarking

2.1. Generative adversarial network (GAN)

GAN is a sort of machine learning framework. [38]. Goodfellow et al. proposed the generative model known as GAN [39]. In image processing, the deep learning-related GAN has shown impressive results. An artificial intelligence method, generative adversarial networks, was created to address the generative modeling issue [40]. A generative model looks at a set of training samples and attempts to determine the probability distribution that produced each [41]. A generator and a discriminator are two sub-models found in a GAN that interact and give feedback to one another to create more realistic outputs [42]. The generator is trained to produce samples that resemble the training data set. Inputs are categorized as authentic (from the original training data) or fraudulent by the discriminator. The discriminator seeks to detect fakes, while the generator attempts to replicate reality, as depicted in Fig. 4 [43].

There are several variations of the GAN model, including conditional GAN (CGAN), deep convolutional GAN (DCGAN), laplacian GAN (LapGAN), information maximizing GAN (InfoGAN), etc. [44]. GAN is frequently employed in watermarking for both watermark generation and verification. Some of the research publications about image

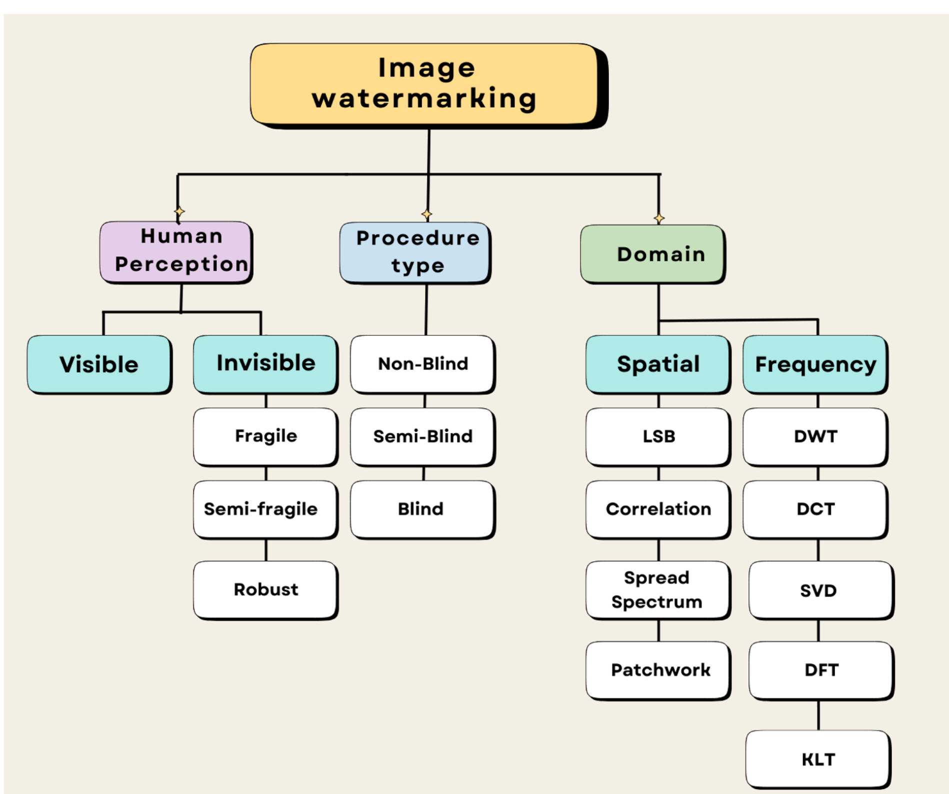

Fig. 1. Digital image watermarking classification.

watermarking-based GAN are illustrated below, and the summary of the GAN model in these research papers is illustrated in Table 1. Singh et al. [45] propose an image watermarking application for healthcare scenarios using the GAN model. The medical image is first subjected to randomized singular value decomposition (RSVD)–based encryption and a 3D chaotic map before being watermarked. GAN is used to create a final mark by embedding the patient's identifying number and the hash of the medical picture into the hospital logo image. RSVD-based embedding and redundant discrete wavelet transform (RDWT) are used to watermark the encrypted medical image. The suggested approach is undetectable and effectively withstands various assaults, according to experimental data.

Annadurai et al. [46] suggest a convolutional generative adversarial neural network method for watermarking digital images. DWT quantization approach based on SVD is used to watermark the image. The proposed technique uses convolutional generative adversarial neural networks to segment and classify watermarked data. The test results show how resilient the recommended watermarking method is against re-encoding attacks and how strong it is against image handling attacks. Regarding training accuracy, the analysis yielded 98%, validation accuracy of 95%, normalized correlation (NC) of 71%, and SSIM of 56%.

Pang et al. [47] suggested a novel GAN deep learning model-based blind watermarking open-source dataset protection technique. The blind watermarking technique applies invisible watermarks to some images in the data collection. This approach suggests using UNet in addition to GAN. When required, the watermark is extracted to fulfill the verification function when there are suspicions that a dubious model was improperly trained.

Huang et al. [48] provide a robust, attention-guided image watermarking model based on GAN. Employ the FFM to acquire multi-layered and deep image features for watermark fusion to avoid the disadvantage of extracting specific image features. An attention module (AM) reduces image distortion brought on by watermark embedding. The discriminator is utilized to recognize the encoded picture from the original image to increase the visibility of watermarks.

2.2. Deep neural network (DNN)

DNN is a neural network with specific intricacy [49]. It is comparable to stacked neural networks, which have many layers—typically two or more—input, output, and at least one hidden layer situated in between, as depicted in Fig. 5 [50]. Various layers, including convolutional, max-pooling, dense, and other special layers, make up the DNN hidden layer structure [51]. With these extra layers, the model can better comprehend issues and offer the best solutions for intricate tasks [52].

Some of the research publications about image watermarking-based DNN are illustrated below, and the summary of the DNN model in these research papers is illustrated in Table 2. Ge et al. [53] proposed a DNN-based document image watermarking method. The watermark is intended to be embedded and extracted using an encoder and a decoder. A noise layer has been created to simulate the many types of assaults that might occur. A text-sensitive loss function is applied to limit embedding changes to characters. Although the PSNR and SSIM appear to be adequate, the watermarked image's visual impact is unsatisfactory, likely because the document image's backdrop is visible. In light of this, they suggested an embedding strength modification technique to improve image quality while maintaining extraction accuracy.

Jaiswal et al. [54] proposed a DNN method for blind color image watermarking. The input image is transformed using integer wavelet transform (IWT), and feature reduction is achieved by principal component analysis (PCA). Watermark images are encoded in the blue color channel. Watermark extraction is done using a binary classification technique. The DNN is trained on a 512 × 11 training set and a 512 × 10 testing set. Sinhal et al. [55] suggest a blind and robust medical image watermarking technology based on DNN. The LZW (Lampel-Ziv-Welch) compression method is used during watermarking to maintain the original picture's region of interest (ROI) data. IWT is used for embedding the watermark. The hash keys for the ROI and RONI (region of non-interest) areas are produced using the SHA-256 technique. Robust watermark extraction is done using a deep neural network-based architecture.

Battarusetty et al. [56] provide a novel method for utilizing DNN with a wavelet-based technique to create watermarked images with increased invisibility. The locations where the watermark should be embedded are selected using the modified DNN. The Squirrel Search–-Grey Wolf Optimisation (SS–GWO) optimizes the DNN training method. Adaptive DWT is used for watermark embedding. Wang et al. [57] provide a reversible image-concealing network-based DNN model watermarking system to safeguard DNN models' copyrights. The suggested solution successfully protects DNN model copyrights while maintaining the key functionality of the host models. The test outcomes show that the proposed framework can provide accuracy and efficacy. The model is applied to five datasets and 11 prominent DNN models.

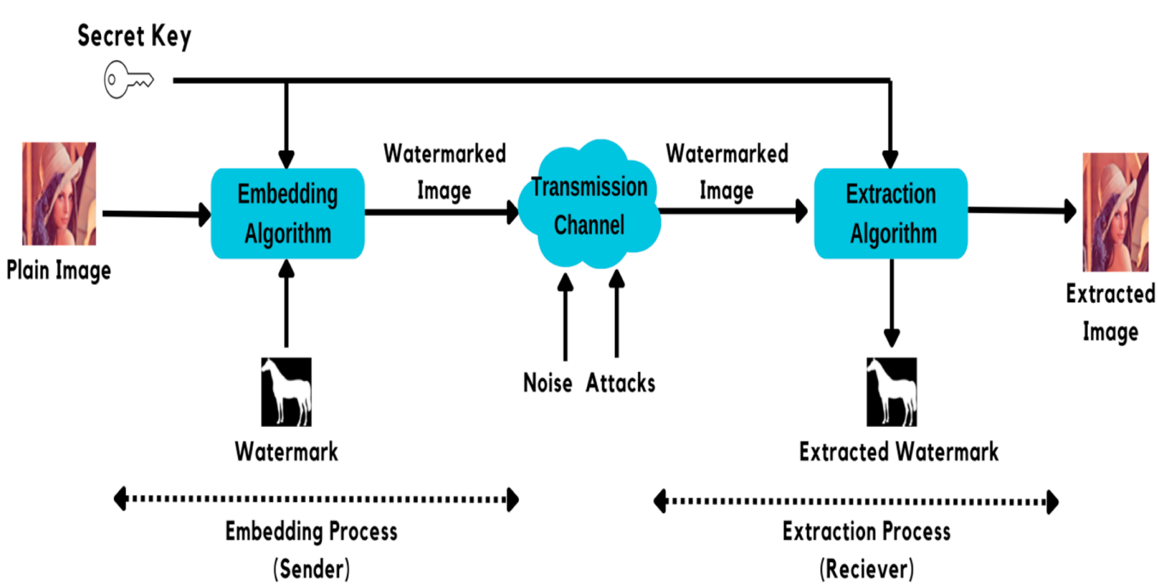

Fig. 2. Digital image watermarking procedures.

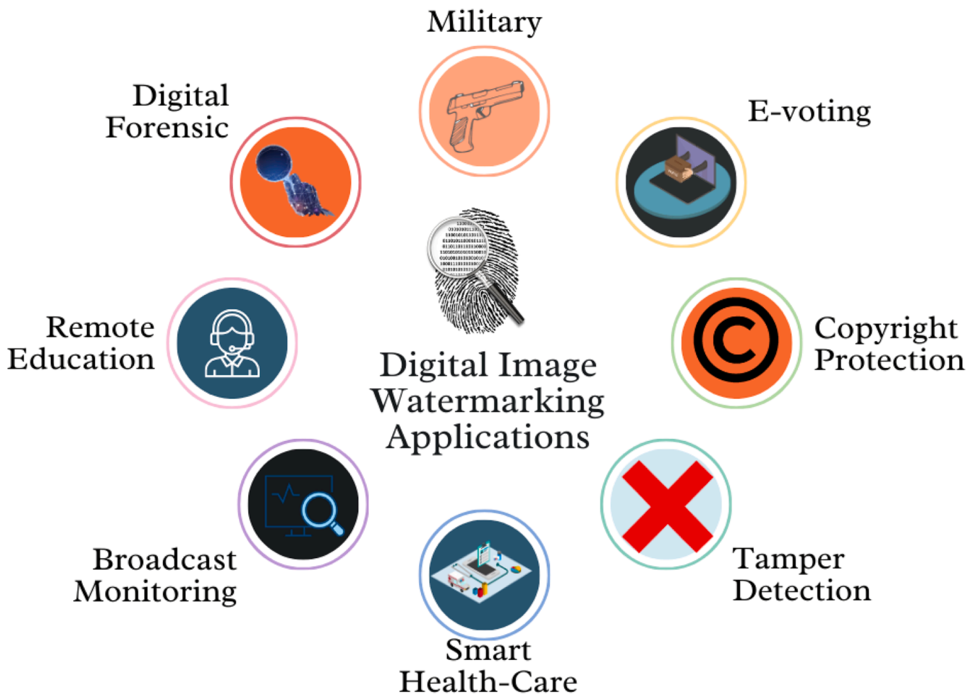

Fig. 3. Digital Image Watermarking Applications.

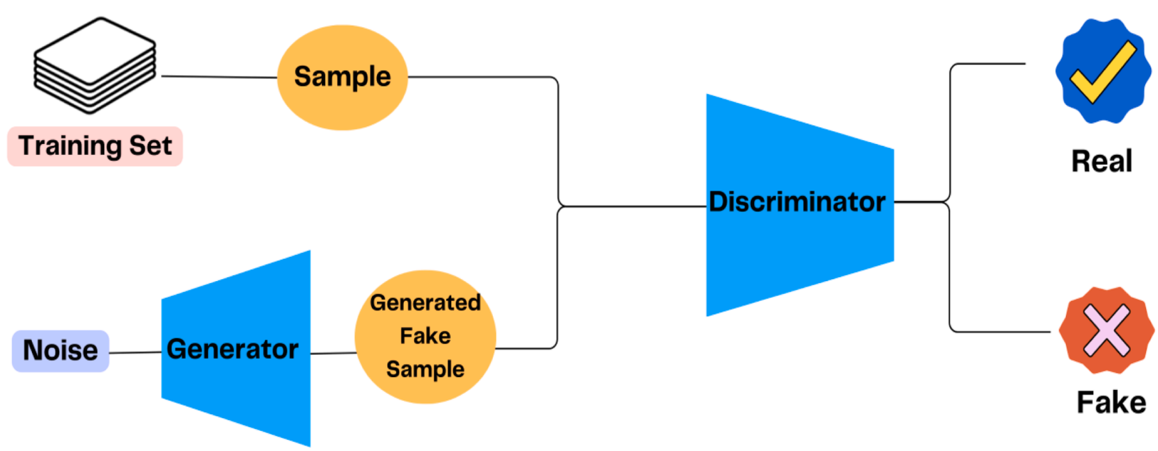

Fig. 4. Illustration of how GAN works.

Table 1
Summary of existing GAN model research papers.

Reference number	Ref. [45]	Ref. [46]	Ref. [47]	Ref. [48]
GAN Role	Secure watermark generation	Segmentation and classification of watermarked data	enhanced feature sharing UNet with GAN is designed as the extraction network	The encoder-noise-decoder structure is used for end-to-end training.
Objective	prevent data leakage in healthcare scenarios	assure the robustness and security of watermarking	overcome the drawback of extracting limited image features	protection algorithm suitable for multiple paired open-sourced datasets
Working Strategies	Chaotic map, RSVD	SVD-DWT,	feature fusion module, attention module	UNet with GAN
Domain	Frequency domain	Frequency domain	Trained DL network	Trained DL network
Robustness Evaluation	NC=0.9999, and BER = 0	BER= 71%, and NCC=71%,	bit accuracy (BA) is used to validate watermarking robustness under five types of noises	BER=1.748%,
Invisibility	PSNR=48.56 dB, SSIM=0.9997	SSIM=56%,	PSNR=41.48, SSIM=0.99000	PSNR=28.537, SSIM= 0.856

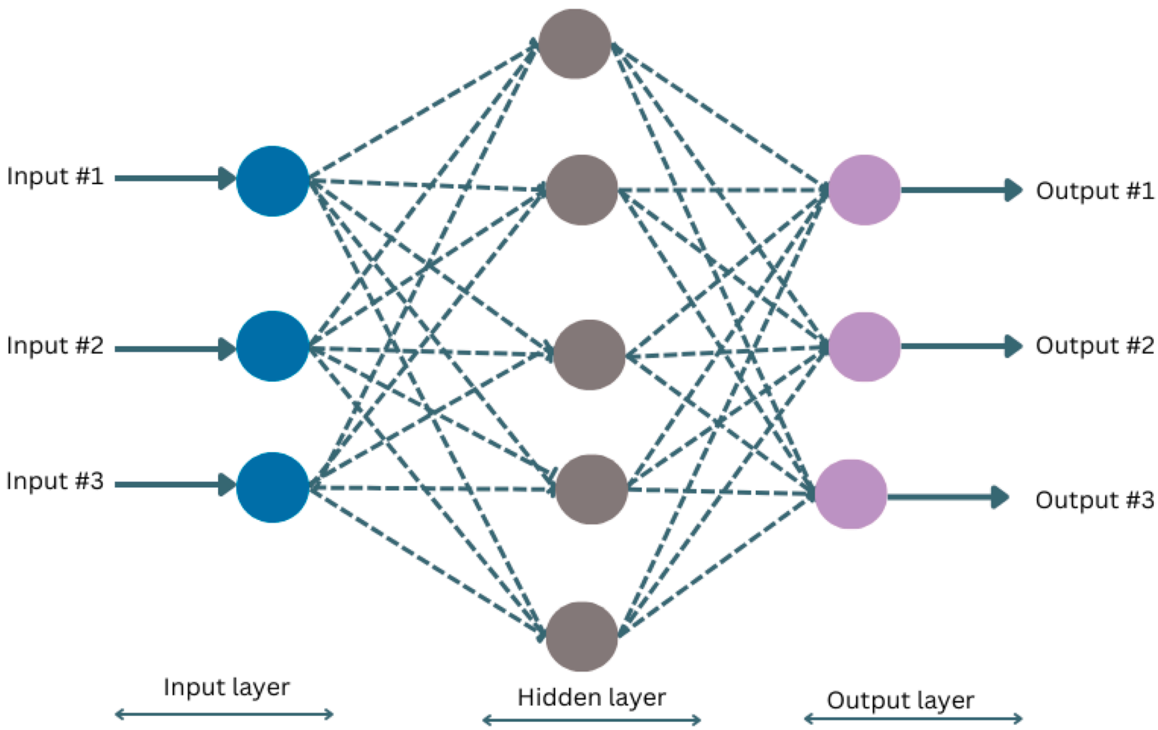

Fig. 5. DNN architecture.

Table 2
Summary of existing DNN model research papers.

Reference number	Ref. [53]	Ref. [54]	Ref. [55]	Ref. [56]	Ref. [57]
DNN Role	Embed and extract the watermark.	Watermark extraction	Watermark extraction	Select the regions for embedding the watermark	Model watermarking and verification
Objective	Watermarking scheme based on document image	Providing a better trade-off between imperceptibility and robustness	Provide effective security solutions for medical images.	Offer an improved watermarking approach	protect the copyrights of DNN models
Working Strategies	Encoder-decoder-noise layer	IWT, PCA	LZW, IWT, SHA-256	SS-GWO, ADWT	DNN model
Domain	Trained DL network	Frequency domain	Frequency domain	Frequency domain	Trained DL network
Robustness Evaluation	bit accuracy is used to validate watermarking robustness under six distortions	Average NC= 0.9953,	BER=0, NC=1	NC=0.93709	—
Invisibility	PSNR=41.07 dB, SSIM=0.965	Average PSNR= 35.57 dB, SSIM=0.9969	PSNR=40.27, SSIM=0.9445	PSNR=19.804	PSNR reaches a maximum of 43.10, and SSIM reaches a maximum of 0.9932.

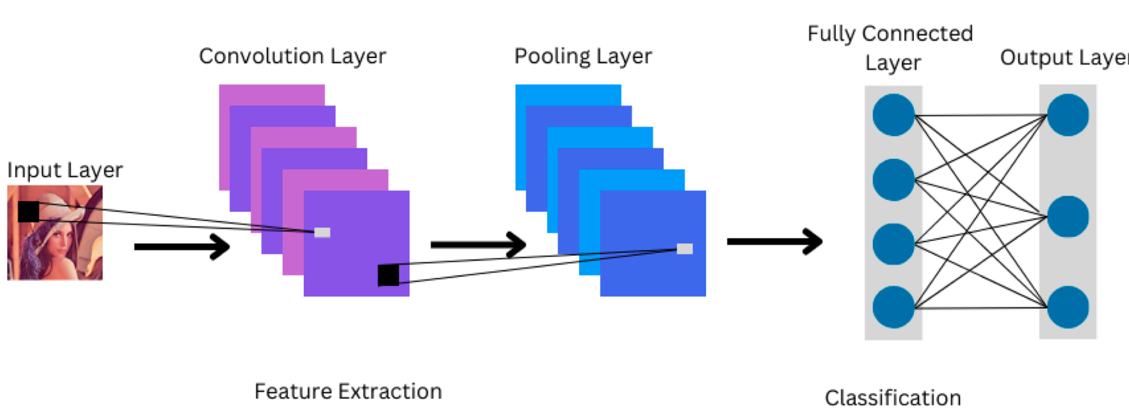

Fig. 6. CNN architecture.

2.3. Convolution neural network (CNN)

One kind of deep learning neural network architecture is CNN. CNN handles grid-like data structures found in images [58]. CNN, particularly well-suited for image analysis, has been used in many contexts as object recognition and image classification [59]. Three layers are commonly seen in a CNN, as depicted in Fig. 6: the convolutional layer, the pooling layer, and the fully connected layer [60].

The convolution layer is the CNN's most essential component. It carries the majority of computational strain on the network [61]. The Pooling layer decreases the spatial size of the Convolved Feature. This process reduces the computing resources needed for dimensionality reduction in the data processing [62]. Pooling comes in two varieties: max pooling and average pooling. The largest value from the area of the picture that the Kernel covers is returned by max pooling [63]. Conversely, Average Pooling yields the mean of all the values from the area of the picture that the Kernel covers [64]. The representation between the input and the output is mapped with the aid of the fully connected layer [65].

Different CNN architectures are available, such as [66]:

- LeNet
- AlexNet
- VGGNet
- GoogLeNet
- ResNet
- ZFNet

Several research publications about image watermarking-based CNN are illustrated below, and the summary of the DNN model in these research papers is illustrated in Table 3.

Liu et al. [67] suggest a watermarking method that uses CNN to create a zero watermark image by superimposing the host image's color style over the watermark logo's information. A styled picture is created by iterating between the host image's style and the content of a watermark logo with a timestamp. The styled picture is encrypted using Arnold transform. Convolutional neural networks are used to verify the copyright of images. Their loss function differs from the original watermark logo and the picture produced. Experiments indicate that the suggested zero-watermarking approach is extremely resistant to standard image processing and geometric assaults.

Chacko et al. [68] suggest a novel method of embedding technology that depends on several embedding parameters to watermark images. Robust watermarking is achieved by defining a novel approach that uses superpixel, DCT, and Harris Hawks Optimisation (HHO). The watermarking bits have been merged with a deep learning convolution neural network (DLCNN) using integration standards. A DLCNN-based pattern recognition model is used to find and extract watermarks, which are then recognized using the HHO method. The suggested technique outperforms previous PSNR, BER, and NCC methods.

Wu et al. [69] present a recovery watermarking approach utilizing convolutional networks and DWT that is fragile. The watermark is based on a downsampled image of the source image. The watermark is inserted in DWT coefficients. The watermark is utilized for the detection and recovery. A CNN is proposed for recovering tampered images. The experimental findings suggest that the proposed technique detects tampering more effectively.

Singh et al. [70] Utilise CNN to develop a novel watermarking method for digital images. The encoder network extracts latent characteristics from cover and secret images, which are then concatenated to create a watermarked image. The recipient uses a denoising autoencoder network to eliminate noise from the picture, followed by a CNN to extract the secret mark. Darwish et al. [71] suggest a method for zero-watermarking color images with CNN and 2D logistic-adjusted Chebyshev map (2D-LACM). Deep feature maps are extracted using the pre-trained VGG19 from the original image. The feature maps were combined into a featured picture, and the owner's watermark sequence was added with an XOR operation. 2D-LACM ensures security by encrypting the watermark and scrambling the feature matrix. Experimental findings indicate that the suggested approach is both invisible and resilient. Hsu et al. [72] present a high-capacity QR decomposition-based blind watermarking technique for color photos using AI technology. The technique employs particle swarm optimization (PSO) and super-resolution convolutional neural networks (SRCNN). PSO optimizes parameters for imperceptibility and resilience. SRCNN refines derived watermark images for better identification. The suggested approach is very resilient against many assaults, except JPEG compression.

Mahto et al. [73] create a durable and imperceptible watermarking technique using hybrid optimization. The technique uses spatial and transform domains to embed hidden markings into cover material. The multi-type mark adds legitimacy to the suggested algorithm by using a proper scaling factor. The watermarked image is encrypted with an enhanced encryption method. A denoising CNN is used to increase the algorithm's resilience. Barlaskar et al. [74] use a DWT-DCT framework to develop a blind watermarking system with an optimum gain factor for embedding a binary watermark in any host picture. DCNN extracts features that aid the support vector regression (SVR) in estimating geometric assaults. A statistical non-geometric correction approach retrieves the watermark under different non-geometric attack scenarios.

Zhang et al. [75] introduce CNN and DCT to provide a blind resilient image watermarking system. A channel weighting module using a modified Gaussian distribution is developed to adjust watermarking information distribution and intensity. A spatial weighting module to enhance the visual quality of watermarking is built. A frequency domain channel-attention enhancement module improves watermarking resilience by sensing and enhancing channel signals. Ingaleshwar et al. [76] present a WCFOA-based Deep CNN technique for embedding hidden messages into cover images. The proposed WCFOA combines the Water Wave Optimization and Chaotic Fruit Fly Optimization techniques. After extracting features, the Deep CNN classifier trained by the proposed WCFOA determines the best place to implant the secret messages. The wavelet transform is used for embedding.

Rai et al. [77] proposed an enhanced chimp optimization (ECO) technique that uses deep fusion CNN to provide robust watermarking. This approach uses an embedding and extraction network for watermarks. The embedding network's octave convolutional model reduces spatial redundancy while capturing several features. The extraction network's pyramid feature extraction module extracts local features, while dilated convolutions reduce model parameters. Tavakoli et al. [78] suggested using CNN and wavelet transformations to create a watermarking network for embedding and extracting the watermarks. The network can take all types of watermarks, is agnostic of host picture resolution, and has just 11 layers due to wavelet transform. Performance is evaluated using the extracted watermark's similarity to the original and the host image's similarity to the watermarked one.

Guo et al. [79] proposed a blind watermarking system based on deep learning. The suggested method utilized a modified CNN to extract watermarks with a block classification approach. The modified CNN creates a residual module with fewer batch normalizations, resulting in faster watermarking. The suggested approach is effective against typical signal processing assaults, as demonstrated by experimental findings. Geng et al. [80] study the confluence of deep learning with blockchain technologies in copyright protection. A digital picture watermarking model is designed, and SVD and deep learning approaches are used to create a flexible watermarking system. The blockchain model's performance parameters use a practical byzantine fault tolerance consensus method for consistent and reliable transactions. The algorithm's performance was examined using normalized cross-correlation (NC), and attack tests were conducted on original photos to evaluate its resilience and practicality.

Bai et al. [81] present an infrared images steganography method

Table 3
Summary of existing CNN model research papers.

Reference number	CNN Role	Objective	Working Strategies	Domain	Robustness Evaluation	Invisibility
Ref. [67]	Watermark extraction to verify the image copyright	construct the zero-watermark image by addition rather than XOR operation	Arnold transform	Spatial	correlation coefficient under various attacks is computed	Correlation coefficients for ten watermark logos are computed
Ref. [68]	Identify and extract the watermark.	Efficient positioning and parameter optimization of image watermarking	DCT, Arnold transform, HHO	Frequency domain	NC and BER under different attacks are calculated	PSNR= 46 dB
Ref. [69]	Recovery of the tampered image	Image authentication and tamper recovery.	DWT	Frequency domain	___	PSNR around 41.87 dB and SSIM around 0.977
Ref. [70]	Watermark embedding and extraction	Train DL network to reduce human intervention	Encoder-decoder network	Trained Dl network	NC=0.9996	PSNR=44.48 dB, SSIM=0.9997
Ref. [71]	Deep feature extraction	Improved zero-watermarking resistance	2D-LACM	Trained DL network	BER below 0.0044, NC above 0.9929	Average PSNR=33.1537
Ref. [72]	Refine the extracted watermark image.	solves the shortcomings of QRD-based watermarking technology	PSO, SRCNN	Frequency domain	BER=0.107, NCC=0.89	PSNR=38.02, MSSIM=0.915
Ref. [73]	The denoising process of the recovered watermark image	Enhanced watermarking using hybrid optimization	LWT-Schur-T-SVD, HPSOF, SIE scrambling	Hybrid domain	NC and BER are approaching 1 and 0, respectively,	The average PSNR is 57.7124 dB,
Ref. [74]	Watermarked image deep feature extraction.	A genetic algorithm has been used for optimizing watermark embedding	DWT-DCT, SVR,	Frequency domain	NC~0.9784	The average PSNR is 47.81 dB
Ref. [75]	Watermark embedding and extraction	control the distribution and intensity of watermarking information	2D-DCT	Hybrid domain	BER is computed under some real-world attacks	PSNR and SSIM are computed under some real-world attacks
Ref. [76]	Feature extraction and optimal region selection	Robust optimized watermarking method	WCFOA, DWT	Frequency domain	NC=1	PSNR=45.2157
Ref. [77]	Feature extraction	Apply optimization algorithm based on CNN for better watermarking	ECO, OCM, FFN	Hybrid domain	NC=0.98, BER=0.038	PSNR=54.64, SSIM=0.989
Ref. [78]	Watermark embedding and extraction	Improve watermarking accuracy and performance	DWT, Haar wavelet	Frequency domain	BER= 0.0003	PSNR=40.1
Ref. [79]	Watermark embedding and extraction	Improve watermarking performance	Stochastic Gradient Descent(SGD)	Spatial domain	NC and BER values are computed under some attacks	PSNR and SSIM values are computed under some attacks
Ref. [80]	Feature extraction	fusion of DL and blockchain technology	SVD, blockchain	Frequency domain	NC values are computed under some attacks	__
Ref. [81]	Feature extraction- Image pre-processing (SRCNN)- improve the accuracy of the pixel prediction (CNNP)	Steganography infrared images	SWT, SRCNN, CCNP	Hybrid Domain	__	Average PSNR=59.27 dB for 10,000 bits
Ref. [82]	Feature extraction	Propose a multi-watermarking technique	DCT, logistic map, and hash function	Frequency domain	NC under different attacks is calculated	PSNR under different attacks is calculated
Ref. [83]	Feature extraction					
Ref. [84]	Geometry attack classification and correction procedure	balancing the imperceptibility and robustness by optimization	MRFO, DTCWT, SVD	Frequency domain	Average BER=0.067, NCC=0.841 without geometric correction	Average PSNR=54.721
Ref. [85]	Feature extraction	robust medical watermarking algorithm	chaotic Henon mapping	Trained DL network	NC under different attacks is calculated	PSNR under different attacks is calculated
Ref. [86]	Watermark embedding and extraction	watermarking scheme for high-definition images	Embedding -revealing network	Trained DL network	NC=0.9685, BER=0.0343	PSNR=38.8790, SSIM=0.9581
Ref. [87]	Selecting the optimal embedding regions	innovative image watermarking approach	HWT, CBEO	Frequency domain	Max. NC=0.761, least BER=0.047	Max. PSNR=24.989 dB, Max. SSIM=0.969
Ref. [88]	Feature extraction	Overcome geometric attacks	DWT, DCT, chaotic scrambling	Frequency domain	PSNR is computed under various attacks. Most NC larger than 0.50	PSNR is computed under various attacks
Ref. [89]	Image denoising and super-resolution reconstruction	Directly embeds color pixel values into the designated DCT coefficients.	DCT	Frequency domain	Average ZNCCs=1 under no attack	PSNR=37.253, SSIM=0.961
Ref. [90]	Feature extraction- zero-watermark verification	replace the traditional XOR operation with zero watermarking	The generator network and the detector network	Trained DL network	NC is computed under various attacks	PSNR is computed under various attacks
Ref. [91]	Feature extraction	Improve medical watermarking robustness	DCT	Frequency domain	NC is computed under various attacks	PSNR is computed under various attacks
Ref. [92]	Feature extraction	Improve medical watermarking robustness	DCT, chaos map, Arnold transform	Frequency domain	NC is computed under various attacks	PSNR is computed under various attacks
Ref. [93]	Processing the watermark of images	Improve medical watermarking robustness	DWT, DCT, logistic map	Frequency domain	NC is computed under various attacks	__
Ref. [94]	Feature extraction	Improve better robustness	DCT, Tent mapping	Frequency domain	NC is computed under various attacks	PSNR is computed under various attacks
Ref. [95]	Feature extraction	authenticating and securing healthcare records	NSST, SVD, SSFC	Frequency domain	NC is computed under various attacks	PSNR is computed under various attacks

using smooth wavelet transform (SWT) and CNN. Half of the input infrared picture is pre-processed using SRCNN and SWT algorithms. The remaining half of the infrared picture is then predicted using the convolutional neural network predictor (CNNP). The suggested CNNP model incorporates an attention strategy to enhance prediction accuracy. This concept does not need costly devices or extensive storage space for training. Fan et al. [82] proposed an image watermarking technique that combines CNN Inception V3 and DCT for image processing and feature extraction. This approach uses a logistic map and hash functions to scramble the watermarks. Three watermark forms were used in this method to test the algorithm's viability and generalizability.

Huang et al. [83] suggest a lossless photo watermarking technique using depthwise overparameterized VGG (DO-VGG) and an attention mechanism. The pre-trained DO-VGG model extracts deep, high-dimensional features from medical pictures. The suggested technique uses deep neural networks to extract features, making it resistant to geometric assaults. Deep-level image features are extracted using CBAM in both channels and spatial dimensions. The suggested approach uses enhanced logistic chaotic mapping and encryption to encrypt the watermarking image. Sharma et al. [84] propose an efficient watermarking technique with great imperceptibility and resilience called MantaRayWmark. The system uses Manta-Ray Foraging Optimization to improve the locally relevant multiple embedding strengths, achieving a compromise between the imperceptibility and resilience of images. A geometric correction method based on CNN technology is proposed to strengthen the watermarking technique against geometric assaults. Zhang et al. [85] present a reliable multi-watermarking solution using GoogLeNet and Henon Map. The GoogLeNet network is trained on a medical training sample set and then used to extract feature vectors from images. Three forms of watermark are inserted and extracted to evaluate the algorithm stability. The feature vectors are combined with the encrypted watermark to complete the zero watermark embedding and blind extraction.

Zhu et al. [86] present a blind image watermarking system using multiscale fusion dilated ResNet (MDResNet) and key-point detection. The watermarking system embeds the watermark in multiple non-overlapping areas, each protected by a private key that refers to different dominating key points. The watermarks are placed in the middle section of the region in such a way that they remain inside the regions even when the picture is geometrically changed. MDResNet embeds and extracts the watermarks. Suresh et al. [87] proposed a deep learning-based chronological bald eagle optimization (CBEO) technique for image watermarking. The host image embeds the watermark by identifying an ideal place using LeNet. The embedding process utilizes the Haar Wavelet Transform (HWT) to enhance resilience. The CBEO method is used to optimize the trainable parameters of the LeNet, which selects the ideal region in the cover image. Nawaz et al. [88] introduced a zero-watermarking system using DWT and ResNet-DCT that can withstand geometric assaults. Deep features are transformed using DWT and DCT, and a perceptual hash function is used to construct the feature vector. ResNet is used to extract deep features in medical images. After obtaining and decrypting the encrypted watermark, the recovered watermark can used to identify the image ownership.

Hu et al. [89] introduced a new approach for blind picture watermarking based on DCT. The watermarking approach utilizes a deep learning network to create high-quality, high-resolution watermarks for visual assessment. ResNet is used for image denoising and super-resolution reconstruction. Xiang et al. [90] introduced a zero-watermark for medical images using style features and residual networks. Deep neural networks obtain high-order statistics from image details and abstract features, making them resistant to geometric assaults. This paper suggests using a residual network for zero-watermark verification instead of the standard XOR procedure. The residual network iteratively learns how to extract the watermark from the zero watermark images and improve its quality with the defined loss function.

Gong et al. [91] suggested a zero watermarking approach using Residual-DenseNet. This approach uses CNN to enhance the zero-watermarking approach by training Residual-DenseNet to extract resilient feature vectors. The chaotic matrix is generated using the logistic map technique for watermark encryption and decryption. This approach performs well against conventional and geometric assaults, as demonstrated by experimental findings. Dong et al. [92] proposed a zero watermarking technique that utilizes an improved NasNet-Mobile CNN with DCT. The watermarked data is twice encrypted using the Chen chaotic system and Arnold mapping technique, improving its security. Migration learning is applied to the upgraded NasNet-Mobile network. The retrieved image features are transformed using DCT to recover low-frequency data, which is then processed using the perceptual hash technique. Finally, medical images are encrypted using zero-watermarking technology.

Nawaz et al. [93] present a zero watermarking approach utilizing DWT, an upgraded MobileNetV2 convolutional neural network, and DCT. An enhanced MobileNetV2 pre-trained network is utilized, followed by transfer learning to extract features from encrypted medical images. A chaotic method is used for encryption, and the generated key is maintained in a remote location to enhance the watermark's security. Li et al. [94] suggested a zero watermarking approach using DCT and an enhanced DarkNet53 CNN. DCT is utilized to extract features from encrypted medical images. DarkNet53 was selected for migration learning for feature extraction from the encrypted images. Encryption is done using Tent Map and Logistic Map.

Anand et al. [95] propose a deep learning-based zero-watermarking approach to safeguarding healthcare records. The watermark image is scrambled to increase security using the step space-filling curve approach. Alexnet extracts features from host images that are visibly marked. Zero watermarking is implemented using non-subsampled shearlet transform (NSST) and SVD.

2.4. Other deep learning models-based image watermarking

Rai et al. [25] offered a hybrid approach for picture watermarking that combines fuzzy back propagation neural network (BPNN) with shark smell optimization. Fuzzy-BPNN and the Human Visual System (HVS) assist in identifying places with noise that the human eye may miss. The SSO algorithm determines the ideal embedding positions. The analysis shows that the suggested strategy beats other previous strategies. Liu et al. [96] introduced a new SVD-based image watermarking method. The suggested approach improves image quality by embedding the watermark bit in the first singular value of each color block. Blind watermark extraction uses a generalized regression neural network (GRNN) to avoid keeping the original image information and determine the singular value.

3. Discussion and statistics

Watermarking using deep learning has lately acquired popularity due to its wide range of applications and considerable influence on image content security. Deep learning can select the optimum embedding places in the cover image, reduce noise in the final watermark, and simulate assaults to extract watermarks. Deep learning is now used in smart city applications. Smart city researchers have used deep learning in various applications, including traffic prediction, healthcare, air quality prediction, and public safety. Most watermarking systems face the challenge of balancing embedding capacity, imperceptibility, and robustness. Combining encryption with watermarking enhances watermark security. However, it also increases system complexity.

The various works on using deep learning in image watermarking demonstrated that using deep learning will benefit the digital image watermarking sector. Deep learning models include recurrent neural networks (RNN), BPNN, GAN, DNN, and CNN. This survey contains research articles on image watermarking with deep learning. The study results revealed that three deep learning models are extensively utilized in image watermarking: GAN, DNN, and CNN. This survey studies the three deep learning models extensively employed in digital picture watermarking. As depicted in Fig. 7, CNN is the deep learning model most commonly used in image watermarking.

Many research papers have recently been on deep learning-based image watermarking; this indicates a progressive growth in researchers' interest in this topic. Deep learning is utilized in image watermarking for various tasks such as watermark embedding and extraction, feature selection, and selecting the optimal embedding locations. The study on deep learning-based watermarking technology is still in its early stages. Even if there isn't much research, they have increased recently. As depicted in Fig. 8, the number of research papers utilizing CNN in image watermarking increases.

Our survey provided many innovative research publications for deep learning-based image watermarking. Neural networks are used to embed and remove watermark information to maintain invisibility and increase robustness against attacks. The most problematic aspect of watermarking models is the trade-off between invisibility and robustness. A speedier watermarking model with fewer processing resources is preferred for efficiency, but it must also be resistant to many types of attacks. CNN is the most often used neural network for image watermarking due to its superior performance. Although CNN models might suffer from several limitations as follows:

- The train and process of a CNN model required a larger dataset.
- Misconfigured network settings might lead to longer training periods.
- Operations such as max-pool enhance latency in CNN models.
- Overfitting and underfitting are common issues with CNN networks because of their complexity.

Deep learning has emerged as one of artificial intelligence research's most popular and fascinating areas. It is a subfield of machine learning that analyses huge datasets using neural networks. In recent years, deep learning has received a lot of interest in the image processing sector. Like any other technology, deep learning offers benefits and drawbacks, as depicted in Fig. 9. We will discuss the benefits and drawbacks of deep learning.

Advantages of deep learning-based image watermarking:

1. Automatic feature extraction eliminates the need for manual intervention.
2. Deep learning dynamically extracts high- and low-level features from large datasets.
3. Deep learning algorithms can process structured and unstructured data like images, audio, and texts.
4. The same neural network-based method may be used for various applications and datasets.
5. Handling massive and complex datasets
6. GPUs can do massively parallel computations, which are scalable for big amounts of data.
7. Deep learning models are extremely adaptive. The models can be fine-tuned or tailored to new tasks.
8. Deep learning can find non-linear relationships in data.
9. The deep learning architecture is adaptable, allowing it to be used to solve future issues.

While deep learning has surpassed traditional machine learning approaches, it is unsuitable for all applications. Researchers must consider a variety of restrictions:

Disadvantages of deep learning-based image watermarking:

1. Deep learning requires a vast amount of data to perform better than other techniques, which requires a lot of time and resources for data collection.
2. Training a particular model with large datasets requires more processing resources for extensive computing requirements than other machine learning models.
3. Deep learning models might be challenging to understand. The deep learning applications act as a black box that converts inputs to outputs, making it unsuitable for people who want to understand

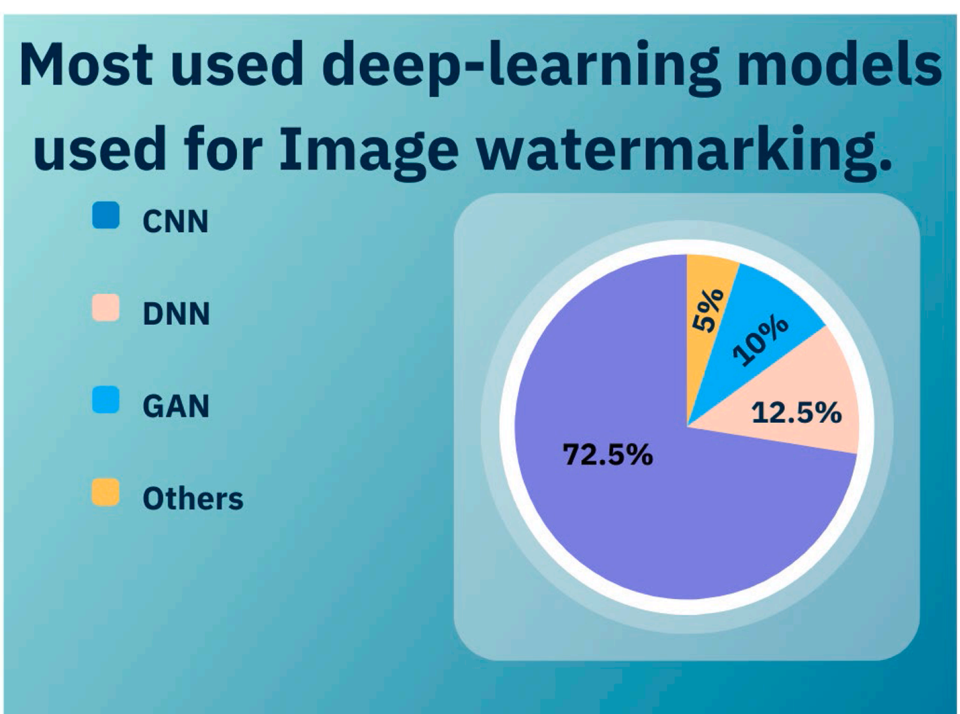

Fig. 7. Most used deep-learning models for image watermarking.

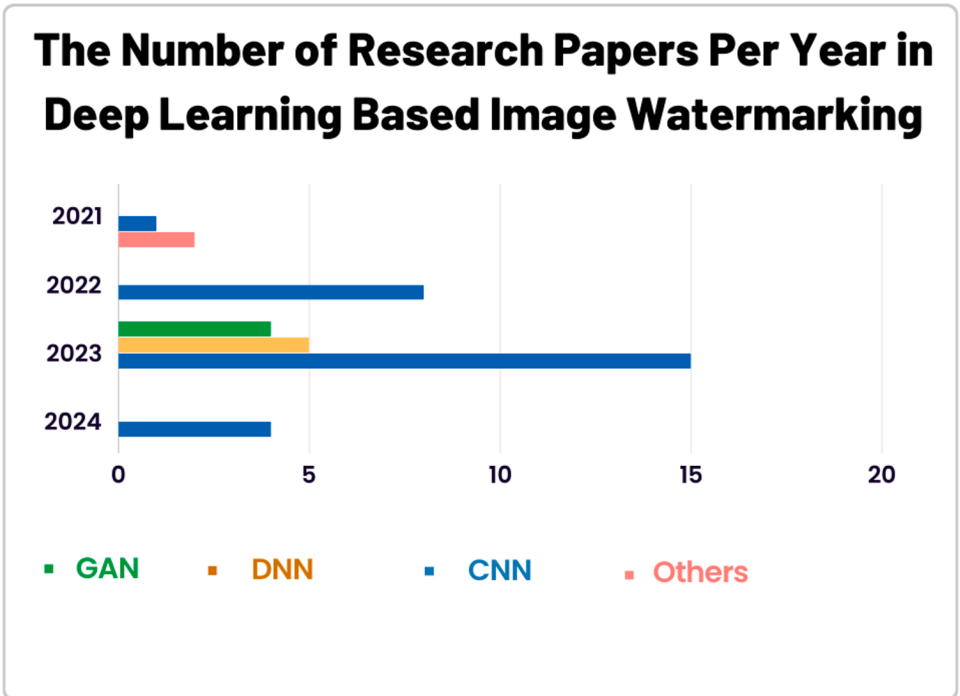

Fig. 8. Number of research papers per year in deep-learning-based image watermarking.

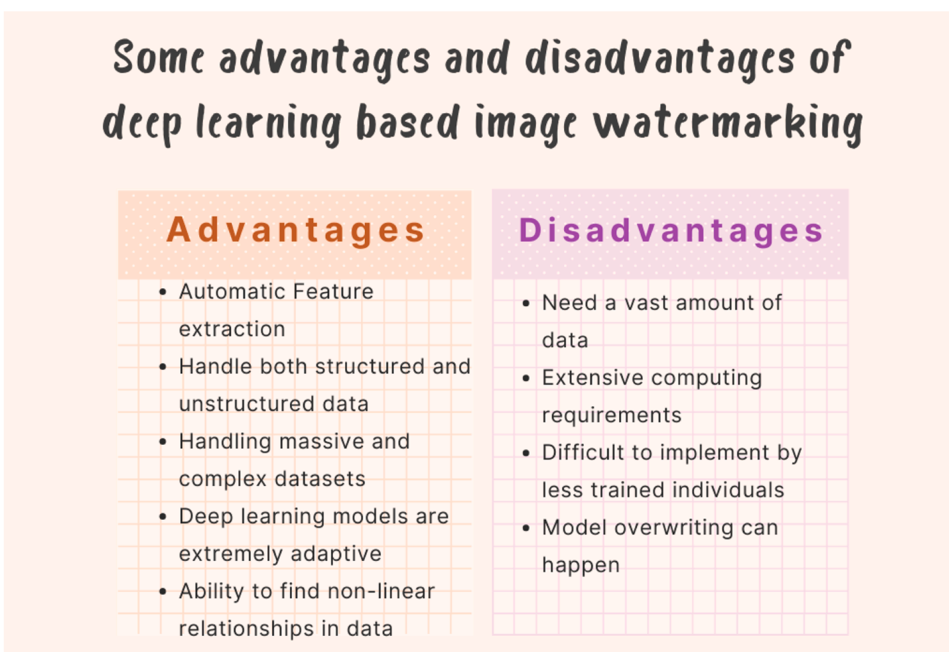

Fig. 9. Some deep learning advantages and disadvantages in image watermarking.

application decision-making procedures. As a result, it isn't easy for less trained individuals to implement it.

4. Pre-trained models are commonly employed to reduce training complexity, and this can cause model overwriting.
5. The robustness of watermarking is dependent on the number of samples used during training and the loss function utilized.

Therefore, further study is required to fully cover the limitations of deep learning for image watermarking applications.

4. Future directions

Image watermarking-based deep learning is a new and emerging research field with numerous opportunities for advancement. The recent proliferation of research articles on deep learning-based picture watermarking shows that interest in this field has steadily increased. Here, we outline our recommendations for potential future research areas:

- The deep watermarking models addressed in this review are resilient and effective for image watermarking and may be applied to various media types because this is a new research topic.
- Safeguarding machine learning models as intellectual property is also a significant research topic for digital watermarking.
- Compared to spatial domain-based methods, frequency domain-based watermarking techniques demonstrate higher durability. We can incorporate multiple frequency domains into the watermarking technique to improve security.

5. Conclusion

Image watermarking has lately acquired popularity due to the rise of copyright concerns. Several watermarking approaches have been developed to improve image quality and resilience against various assaults. Deep learning improves computer vision by fitting and generalizing complicated features. Watermarking using deep learning has lately attracted substantial interest due to its various widely used applications and a major influence on image security. Deep learning enables adaptive image watermarking by dynamically extracting high- and low-level features from large datasets. This survey thoroughly examines various research publications on digital image watermarking in deep-learning environments. We reviewed the watermarking idea, steps, performance measurements, and applications and illustrated the deep learning definition and the different types of neural networks. Finally, we highlighted outstanding issues and future research paths for image watermarking with deep learning.

Declaration of competing interest

No conflict of interest exists.

The second author received a fund, which is acknowledged in the manuscript.

The authors understand that the Corresponding Author is the sole contact for the Editorial process. He is responsible for communicating with the other authors about progress, submissions of revisions, and final approval of proofs.

Data availability

Data will be made available on request.

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